# Quantum Simulation of Spin-Dependent Charmonium Spectroscopy with Variational Quantum Algorithms

Master's Thesis

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# Chapter 1

# Introduction

The simulation of quantum systems lies at the heart of quantum computing's promise. As Feynman observed in 1982, classical computers face a fundamental barrier when modelling quantum mechanical systems: the Hilbert space of an $n$-particle system grows exponentially, rendering exact classical computation intractable for even modest system sizes [1]. Quantum computers, by contrast, naturally encode and manipulate quantum states, offering a path toward efficient simulation of physical systems that are otherwise inaccessible [2].

This thesis applies variational quantum algorithms to the spectroscopy of charmonium—bound states of a charm quark and its antiquark—including spin-dependent interactions that produce the experimentally observed mass splittings between states such as the $\eta_c$ and $J/\psi$ mesons. The quarkonium system is physically rich enough to probe the capabilities of near-term quantum algorithms across multiple quantum number channels and excited states, yet small enough to be tractable with only a few qubits.

The remainder of this chapter introduces the three pillars on which this work rests: the fundamentals of quantum computing (Section 1.1), variational quantum methods (Section 1.2), and the quarkonium problem (Section 1.3).

## 1.1 Quantum Computing

### 1.1.1 Why quantum computing?

The classical theory of computation, formalized by Turing and Church in the 1930s, rests on the assumption that information is processed according to classical physics. For decades this abstraction posed no practical limitation. However, the simulation of quantum mechanical systems reveals a fundamental barrier: the state space of an $n$-particle quantum system is a Hilbert space of dimension $2^n$, so that a faithful classical representation requires resources that grow exponentially with system size. A modest system of 50 interacting spin-$\frac{1}{2}$ particles, for instance, demands $2^{50} \approx 10^{15}$ complex amplitudes—

exceeding the memory of any existing classical computer.

In 1982, Feynman proposed a radical alternative: rather than simulating quantum mechanics on a classical machine, one should use a controllable quantum system to simulate another [1]. The exponential scaling that makes classical simulation intractable becomes an *asset*: a quantum processor with $n$ qubits naturally inhabits the same $2^n$-dimensional space as the target system. This observation laid the conceptual foundation for quantum computing.

The subsequent development of quantum algorithms by Deutsch, Shor, and Grover [2] demonstrated that quantum computers are not merely faster classical machines, but represent a genuinely different model of computation. The complexity class $\mathsf{BQP}$ (bounded-error quantum polynomial time) is believed to be strictly larger than its classical counterpart $\mathsf{BPP}$, meaning there exist problems for which quantum algorithms achieve an exponential advantage that no classical algorithm can match [2]. The source of this advantage lies in two uniquely quantum phenomena: *superposition*, which allows exploration of exponentially many computational paths simultaneously, and *interference*, which enables the selective amplification of correct answers and suppression of incorrect ones.

### 1.1.2 Qubits and quantum states

The fundamental unit of quantum information is the *qubit*, a two-level quantum system whose pure state is described by a normalized vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{1.1}$$

where $\alpha, \beta \in \mathbb{C}$ are *probability amplitudes* and $\{|0\rangle\,, |1\rangle\}$ is the computational basis (conventionally identified with the eigenstates of the Pauli $Z$ operator). Unlike a classical bit, which is deterministically 0 or 1, a qubit exists in a coherent *superposition* of both basis states.

Geometrically, any single-qubit pure state can be represented as a point on the *Bloch sphere* via the parameterization

$$|\psi\rangle = \cos\frac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|1\rangle\,, \tag{1.2}$$

with polar angle $\theta \in [0, \pi]$ and azimuthal angle $\phi \in [0, 2\pi)$. The north and south poles correspond to $|0\rangle$ and $|1\rangle$, respectively, while points on the equator represent equal superpositions with varying relative phase.

For a system of $n$ qubits, the state space is the tensor product $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, spanned by $2^n$ computational basis states $|x_1 x_2 \cdots x_n\rangle$ with $x_i \in \{0, 1\}$. A general pure state is

therefore

$$|\Psi\rangle = \sum_{x\in\{0,1\}^n} c_x |x\rangle, \qquad \sum_x |c_x|^2 = 1, \tag{1.3}$$

requiring $2^n$ complex amplitudes to specify. It is this exponential growth of the state space that makes quantum systems both hard to simulate classically and powerful as computational devices.

### 1.1.3 Entanglement

A multi-qubit state is called *separable* (or a product state) if it can be written as a tensor product of individual qubit states, $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$. States that *cannot* be decomposed in this way are *entangled*. The simplest example is the Bell state

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\Big(|00\rangle + |11\rangle\Big), \tag{1.4}$$

which describes two qubits in a joint superposition: neither qubit has a definite value, yet their measurement outcomes are perfectly correlated. This correlation persists regardless of the spatial separation between the qubits and cannot be explained by any local hidden variable theory, as demonstrated by Bell's theorem [3].

Entanglement is not merely a theoretical curiosity—it is a computational resource. Classical correlations can be shared between bits using shared randomness, but entangled qubits carry correlations that are strictly stronger. In quantum algorithms, entangling gates create superpositions over correlated multi-qubit configurations, enabling the exploration of an exponentially large solution space in a way that has no classical counterpart. Without entanglement, a quantum computer can be efficiently simulated classically [2]; it is entanglement that unlocks the full power of quantum computation.

### 1.1.4 Quantum gates and circuits

The time evolution of a closed quantum system is governed by a unitary operator. Quantum computation exploits this by decomposing the desired evolution into a sequence of elementary *quantum gates*: unitary transformations acting on one or two qubits at a time.

The three Pauli operators form a basis for single-qubit operations:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1.5}$$

Together with the identity $I$, they satisfy $\sigma_i^2 = I$, anticommute pairwise ($\sigma_i\sigma_j = -\sigma_j\sigma_i$ for $i \neq j$), and form a basis for the space of $2 \times 2$ Hermitian matrices. Any $2 \times 2$ Hermitian operator—and hence any single-qubit observable—can be written as $A = a_0 I + a_1 X + a_2 Y + a_3 Z$ with $a_i \in \mathbb{R}$.

The Pauli operators generate continuous families of rotation gates on the Bloch sphere:

$$R_\mu(\theta) = e^{-i\theta\,\mu/2} = \cos\frac{\theta}{2}\,I - i\sin\frac{\theta}{2}\,\mu, \qquad \mu \in \{X, Y, Z\}. \tag{1.6}$$

The Hadamard gate $H = (X + Z)/\sqrt{2}$ maps $|0\rangle \mapsto (|0\rangle + |1\rangle)/\sqrt{2}$, creating an equal superposition from a computational basis state.

Two-qubit entangling gates introduce correlations between qubits. The controlled-NOT (CNOT) gate flips the target qubit conditioned on the control qubit being in state $|1\rangle$:

$$\mathrm{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{1.7}$$

The set $\{R_X, R_Y, R_Z, \mathrm{CNOT}\}$ constitutes a *universal gate set*: any unitary on $n$ qubits can be approximated to arbitrary precision by a finite sequence of these gates [2].

A quantum algorithm is specified as a *quantum circuit*: a sequence of gates applied to a register of qubits initialized in a known state, followed by measurement. Figure 1.1 illustrates a minimal circuit that prepares the Bell state of Eq. (1.4) using only a Hadamard gate and a CNOT.

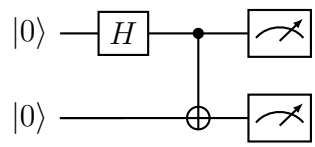


**Figure 1.1:** Quantum circuit preparing the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. The Hadamard gate creates a superposition on the first qubit; the CNOT entangles both qubits. Measurement yields $|00\rangle$ or $|11\rangle$, each with probability 1/2.

### 1.1.5 Quantum parallelism and interference

The mechanism behind quantum computational advantage is often described loosely as "computing on all inputs at once." While evocative, this picture is incomplete: the key ingredient is not parallelism alone, but the ability to make the parallel branches *interfere*.

Consider a unitary $U_f$ that evaluates a function $f : \{0,1\}^n \to \{0,1\}$. Applied to an equal superposition over all $2^n$ inputs, it produces

$$U_f \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle\,|0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle\,|f(x)\rangle\,. \tag{1.8}$$

All function values are encoded in a single quantum state. However, measuring this state yields a *single* random pair $(x, f(x))$—no better than classical evaluation. The power of

quantum algorithms lies in subsequent operations that exploit *interference*: unitary gates are designed so that amplitudes corresponding to desired outputs add constructively, while those corresponding to undesired outputs cancel destructively. The final measurement then yields the correct answer with high probability.

This interplay of superposition, entanglement, and interference is the mathematical core of quantum speedups. Superposition provides exponential parallelism; entanglement enables correlations across qubits that have no classical equivalent; and interference channels the computation toward the correct outcome.

### 1.1.6 Measurement and the Born rule

While unitary evolution is deterministic, extracting information from a quantum system is inherently probabilistic. A *projective measurement* in the computational basis is described by the set of projectors $\{\Pi_x = |x\rangle\langle x|\}_{x\in\{0,1\}^n}$. Given a state $|\Psi\rangle = \sum_x c_x |x\rangle$, the *Born rule* prescribes the outcome probabilities:

$$p(x) = |\langle x|\Psi\rangle|^2 = |c_x|^2. \tag{1.9}$$

Upon obtaining outcome $x$, the state collapses to $|x\rangle$, and any information encoded in the remaining amplitudes is irretrievably lost.

This has a direct practical consequence: a single measurement of $n$ qubits yields only $n$ classical bits, despite the state containing $2^n$ amplitudes. To estimate a continuous quantity—such as an energy expectation value—the measurement must be repeated many times ($N_{\text{shots}}$ repetitions), and the result is subject to statistical uncertainty scaling as $1/\sqrt{N_{\text{shots}}}$.

More generally, to measure an observable $O$ (a Hermitian operator), its expectation value in state $|\Psi\rangle$ is given by

$$\langle O\rangle = \langle\Psi| O |\Psi\rangle = \sum_i \lambda_i p_i, \tag{1.10}$$

where $\lambda_i$ are the eigenvalues of $O$ and $p_i$ the probability of obtaining the corresponding eigenstate. When $O$ is decomposed as a sum of Pauli operators, each Pauli term can be measured independently by rotating to the appropriate eigenbasis before measurement in the computational basis.

### 1.1.7 Density matrices and mixed states

The pure state formalism of Eq. (1.1) describes isolated, perfectly controlled quantum systems. Real quantum processors, however, interact with their environment, leading to *decoherence* and *mixed states* that require a more general description.

The *density matrix* (or density operator) of a pure state $|\psi\rangle$ is

$$\rho = |\psi\rangle \langle\psi| \,. \tag{1.11}$$

A *mixed state*—representing classical uncertainty about which pure state the system is in—is described by a convex combination

$$\rho = \sum_i p_i \, |\psi_i\rangle \langle\psi_i| \,, \qquad p_i \geq 0, \quad \sum_i p_i = 1. \tag{1.12}$$

The density matrix is Hermitian, positive semidefinite, and has unit trace. A state is pure if and only if $\mathrm{Tr}(\rho^2) = 1$; for mixed states, $\mathrm{Tr}(\rho^2) < 1$, quantifying the degree of decoherence.

In the density matrix formalism, unitary evolution becomes $\rho \mapsto U\rho U^\dagger$, and expectation values are computed as $\langle O\rangle = \mathrm{Tr}(O\rho)$. Noise processes—such as the depolarizing channel relevant to this thesis—are described by completely positive trace-preserving (CPTP) maps on $\rho$. For example, the single-qubit depolarizing channel with error rate $p$ acts as

$$\mathcal{E}(\rho) = (1-p)\,\rho + \frac{p}{3}\Big(X\rho X + Y\rho Y + Z\rho Z\Big), \tag{1.13}$$

which with probability $p$ replaces the state with a random Pauli error. This formalism will be central to the analysis of noise effects in Chapter 1 and subsequent chapters.

### 1.1.8 Quantum simulation

Among the many proposed applications of quantum computing—including cryptanalysis, optimization, and machine learning—the simulation of quantum systems stands out as the most natural and arguably the most promising near-term application [1, 4]. The reason is both conceptual and practical: a quantum computer *is* a quantum system, so using it to simulate another quantum system leverages its native capabilities rather than working against them.

The essential mapping is straightforward. A physical system with a $2^n$-dimensional Hilbert space is encoded on $n$ qubits. The system's Hamiltonian $H$ is decomposed into a sum of Pauli operators acting on these qubits:

$$H = \sum_{i=1}^{M} c_i \, P_i, \qquad P_i \in \{I, X, Y, Z\}^{\otimes n}, \tag{1.14}$$

where each $P_i$ is a tensor product of single-qubit Pauli matrices (a *Pauli string*) and $c_i \in \mathbb{R}$ are real coefficients. This decomposition is always possible since the $4^n$ Pauli strings form a complete basis for $2^n \times 2^n$ Hermitian matrices.

Ground state energies, excited state spectra, and dynamical properties can then be

extracted using quantum algorithms that prepare and measure appropriate states on the quantum register. For the small systems considered in this thesis, the encoding requires only 2–3 qubits, but the same framework scales to larger systems where classical methods become intractable.

Quantum simulation of atomic, molecular, and nuclear systems has seen rapid progress in recent years, with demonstrations ranging from small molecules to lattice gauge theories [5]. The quarkonium system studied in this thesis belongs to this broader programme, representing a step toward quantum simulation of QCD bound states.

### 1.1.9 The NISQ era

Current quantum hardware operates in the *Noisy Intermediate-Scale Quantum* (NISQ) regime [4], characterized by processors with tens to hundreds of qubits subject to significant imperfections:

- **Gate errors** on the order of $10^{-3}$–$10^{-2}$ per operation, described by noise channels such as the depolarizing channel of Eq. (1.13);
- **Limited connectivity** between qubits, requiring additional SWAP gates that increase circuit depth;
- **Short coherence times** that bound the number of operations executable before decoherence degrades the quantum state.

These constraints preclude the deep circuits required by fault-tolerant algorithms such as quantum phase estimation, which need thousands of controlled operations and ancilla qubits for error correction.

The NISQ era has therefore driven the development of algorithms specifically designed for shallow circuits and resilient to moderate noise levels. The most successful paradigm to date is the *variational* approach, in which a quantum processor prepares parameterized trial states while a classical computer optimizes the parameters—a hybrid strategy discussed in the next section.

## 1.2 Variational Quantum Methods

### 1.2.1 The variational principle

Variational methods offer a framework for extracting useful results from NISQ hardware. The central idea rests on the *Rayleigh–Ritz variational principle*: for a Hamiltonian $H$ with ground state energy $E_0$,

$$E_0 \leq \langle \psi(\boldsymbol{\theta}) | \, H \, | \psi(\boldsymbol{\theta}) \rangle \equiv E(\boldsymbol{\theta}) \tag{1.15}$$

for any normalized trial state $|\psi(\boldsymbol{\theta})\rangle$ parameterized by classical variables $\boldsymbol{\theta} \in \mathbb{R}^p$. Minimizing $E(\boldsymbol{\theta})$ over the parameter space yields an upper bound on $E_0$, with equality when $|\psi(\boldsymbol{\theta})\rangle$ includes the true ground state in its span.

### 1.2.2 The hybrid quantum–classical loop

In the quantum computing implementation [5, 6], the trial state is prepared by a *parameterized quantum circuit* (also called an *ansatz*) applied to an initial reference state:

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})\,|0\rangle^{\otimes n}\,, \tag{1.16}$$

where $U(\boldsymbol{\theta})$ is a product of parameterized and fixed gates. The energy expectation value is then estimated on the quantum processor. For a Hamiltonian decomposed as a weighted sum of Pauli strings,

$$H = \sum_{i=1}^{M} c_i\, P_i, \qquad P_i \in \{I, X, Y, Z\}^{\otimes n}, \tag{1.17}$$

each expectation value $\langle P_i \rangle$ is estimated independently through repeated state preparation and measurement in the appropriate Pauli basis. The total energy,

$$E(\boldsymbol{\theta}) = \sum_{i=1}^{M} c_i\, \langle P_i \rangle_{\boldsymbol{\theta}}\,, \tag{1.18}$$

is passed to a *classical optimizer* that updates $\boldsymbol{\theta}$ to minimize $E$. This loop repeats until convergence, as illustrated in Fig. 1.2.

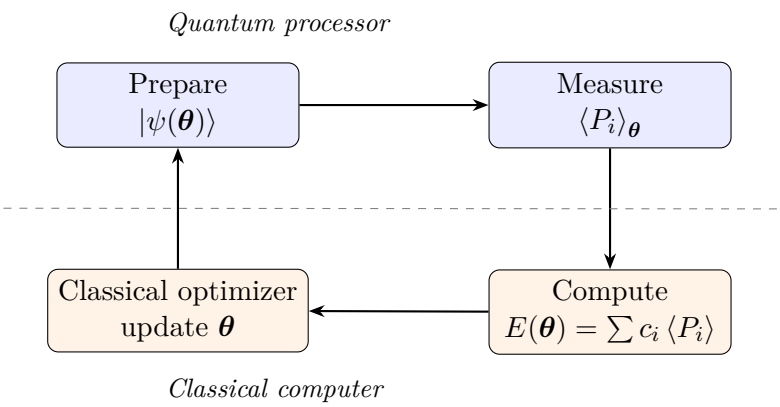


**Figure 1.2:** Schematic of the hybrid quantum–classical variational loop. The quantum processor prepares a parameterized state and estimates Pauli expectation values; the classical computer evaluates the energy and updates circuit parameters. The loop iterates until convergence.

### 1.2.3 Ansatz design and practical considerations

The choice of ansatz circuit is problem-dependent and critical to performance. *Hardware-efficient* ansatze use gates native to the target device, minimizing circuit depth at the cost of physical motivation. *Physically-motivated* ansatze, such as those inspired by unitary coupled cluster theory from quantum chemistry, encode domain knowledge into the circuit structure and may converge faster toward the physical ground state.

The hybrid approach offers several advantages for near-term hardware. The quantum processor handles the exponentially hard task of preparing and measuring many-body quantum states, while the classical optimizer navigates the energy landscape using well-established numerical methods. Circuit depths remain manageable, and variational methods exhibit a degree of inherent noise resilience: small perturbations in the energy surface due to hardware errors are partially absorbed by the optimization [6].

Beyond the ground state, variational frameworks extend naturally to excited states through orthogonality constraints and penalty methods, and to quantum dynamics through variational formulations of time evolution. These extensions broaden the scope of problems addressable on near-term quantum hardware.

## 1.3 The Quarkonium Problem

### 1.3.1 Heavy quark bound states

Quantum chromodynamics (QCD), the gauge theory of the strong interaction, governs the binding of quarks and gluons into hadrons. Among the simplest hadronic systems are *quarkonia*: bound states of a heavy quark and its antiquark ($q\bar{q}$). Charmonium ($c\bar{c}$) systems have served as testing grounds for QCD since the simultaneous discovery of the $J/\psi$ meson in 1974 at BNL [7] and SLAC [8]—an event known as the *November Revolution* in particle physics.

The large mass of the charm quark ($m_c \approx 1.27$ GeV/$c^2$) implies that the quark–antiquark pair moves non-relativistically inside the bound state, with typical velocities $v/c \sim 0.3$. This justifies a potential model approach [9] in which the dynamics are governed by the non-relativistic Schrödinger equation with an effective interquark potential.

### 1.3.2 The Cornell potential

The *Cornell potential* [9],

$$V(r) = -\frac{\kappa}{r} + \sigma\, r, \tag{1.19}$$

provides a successful phenomenological description of the $q\bar{q}$ interaction. It captures the two essential features of the strong force:

- **Short-range Coulomb attraction** ($-\kappa/r$), arising from one-gluon exchange between the color charges, analogous to the electromagnetic Coulomb potential but with QCD coupling;
- **Long-range linear confinement** ($\sigma\, r$), a consequence of the non-Abelian nature of QCD, modelled by a chromoelectric flux tube (string) with tension $\sigma$ connecting the quark and antiquark.

Typical parameter values for charmonium are $\kappa \approx 0.52$ and $\sigma \approx 0.183\ \mathrm{GeV}^2$ [9]. Figure 1.3 shows the characteristic shape of this potential.

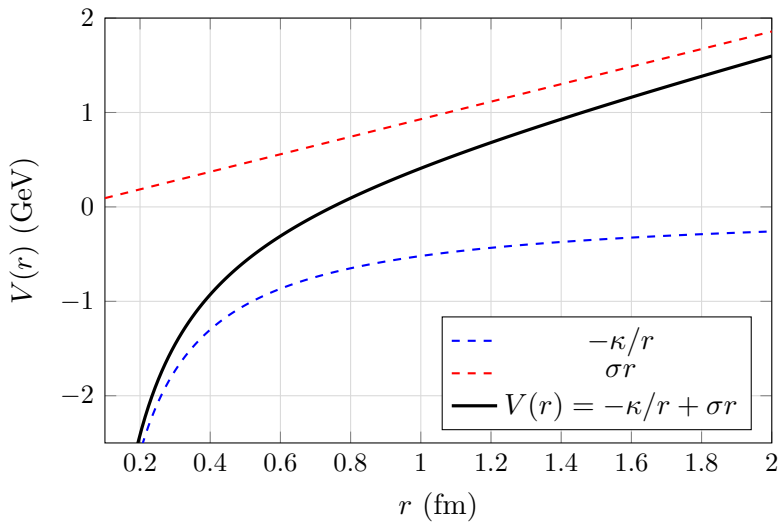


**Figure 1.3:** The Cornell potential as a function of interquark separation $r$. The dashed curves show the Coulomb ($-\kappa/r$, blue) and linear confinement ($\sigma r$, red) components; the solid curve is their sum. At short distances the potential is dominated by one-gluon exchange; at large distances the linearly rising string tension ensures quark confinement.

### 1.3.3 Spin-dependent interactions

The spin-independent Cornell potential alone does not account for the full charmonium spectrum. The mass splitting between states with different spin quantum numbers—for instance, between the $\eta_c$ ($^1S_0$, spin singlet) and the $J/\psi$ ($^3S_1$, spin triplet)—requires spin-dependent corrections. At leading order, the dominant contribution for $S$-wave states is the hyperfine contact interaction [10],

$$V_{\mathrm{hyp}}(r) = \frac{32\pi\,\alpha_s}{9\,m_c^2}\,\delta^{(3)}(\mathbf{r})\,\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}, \tag{1.20}$$

where $\alpha_s$ is the strong coupling constant and $\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}$ is the spin–spin dot product. The eigenvalues of the latter operator distinguish singlet and triplet configurations:

$$\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} = \begin{cases} -\frac{3}{4} & \text{spin singlet } (S = 0), \\ +\frac{1}{4} & \text{spin triplet } (S = 1). \end{cases} \tag{1.21}$$

Quarkonium states are classified in spectroscopic notation ${}^{2S+1}L_J$, where $S$ is the total spin, $L$ the orbital angular momentum ($S, P, D, \ldots$ for $L = 0, 1, 2, \ldots$), and $J$ the total angular momentum. The low-lying charmonium states relevant to this work are:

- ${}^1S_0$: spin singlet, $S$-wave $\longrightarrow$ $\eta_c$ meson ($M \approx 2984$ MeV),
- ${}^3S_1$: spin triplet, $S$-wave $\longrightarrow$ $J/\psi$ meson ($M \approx 3097$ MeV),
- ${}^1P_1$: spin singlet, $P$-wave $\longrightarrow$ $h_c$ meson ($M \approx 3526$ MeV).

### 1.3.4 Quarkonium as a quantum simulation testbed

The full quarkonium Hamiltonian, including kinetic energy and both spin-independent and spin-dependent potentials, reads

$$H = -\frac{1}{2\mu}\nabla^2 + V(r) + V_{\text{hyp}}(r), \tag{1.22}$$

where $\mu = m_c/2$ is the reduced mass. Expanding the radial wavefunction in a truncated basis of $N$ quantum harmonic oscillator (QHO) eigenstates yields a finite-dimensional matrix eigenvalue problem of size $N \times N$, whose entries are determined by the matrix elements of the kinetic and potential operators in the QHO basis.

This matrix can then be decomposed into a sum of Pauli operators (Eq. (1.17)) and encoded on a quantum register of $\lceil \log_2 N \rceil$ qubits, making it directly amenable to variational quantum algorithms. With $N = 3$–$4$ basis states, the problem requires only 2–3 qubits, placing it firmly within reach of current NISQ devices while retaining the essential physics: multiple spectroscopic channels, spin-dependent splittings, and a hierarchy of excited states.

This combination of physical richness and computational tractability makes quarkonium an ideal testbed for benchmarking quantum simulation algorithms against exact solutions and experimental data.

## 1.4 Thesis Overview

This thesis investigates the application of variational quantum algorithms to the charmonium spectroscopy problem described above. We implement and compare several al-

gorithmic approaches for determining both ground and excited state energies across the $^1S_0$, $^3S_1$, and $^1P_1$ channels, assess their performance under realistic noise conditions, and evaluate error mitigation strategies. The goal is twofold: to demonstrate the feasibility of quantum simulation for hadronic physics on near-term hardware, and to provide a systematic comparison of methods that may inform future applications to more complex systems.

The thesis is organized as follows:

- **Chapter 2** develops the theoretical framework, including the Hamiltonian construction in the harmonic oscillator basis and the mapping to qubit operators.
- **Chapter 3** presents the quantum algorithms employed, covering both eigenvalue-based and time-evolution-based variational approaches.
- **Chapter 4** details the numerical results: energy spectra, convergence properties, and cross-method comparisons.
- **Chapter 5** addresses the effects of hardware noise and discusses error mitigation techniques.
- **Chapter 6** concludes with a summary of findings and an outlook on future directions.

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